Task 2 – SQL Database Setup and Query Analysis Using Dataset 1

Step 1: Database and Table Setup

Objective: To verify the database and the structure of the tables created for the space-mission dataset.

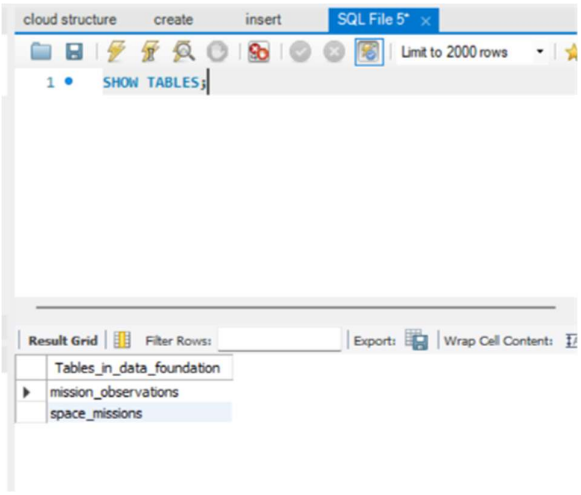
Database Name: data_foundation

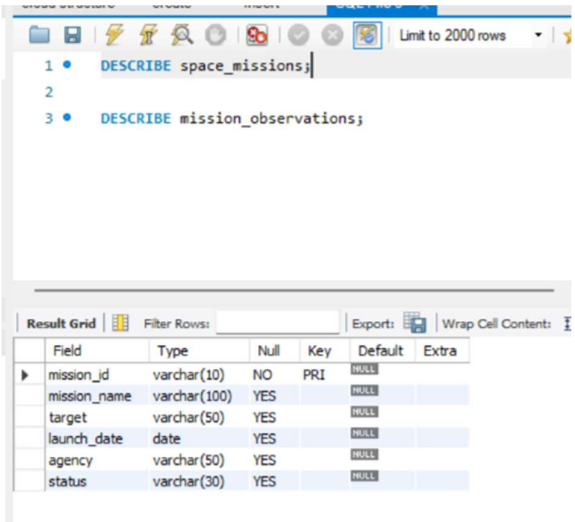
Tables Used:

1. space_missions — stores mission details, including mission name, target, launch date, agency, and status.
2. mission_observations — stores observation details, including region, distance, signal strength, and data collected.

SQL Concepts Used: USE, SHOW TABLES, and DESCRIBE.

Observation: The database and table structures were checked in MySQL Workbench.





Step 2: Data Loading and Verification

Objective: To verify that the space-mission dataset has been loaded correctly into the database.

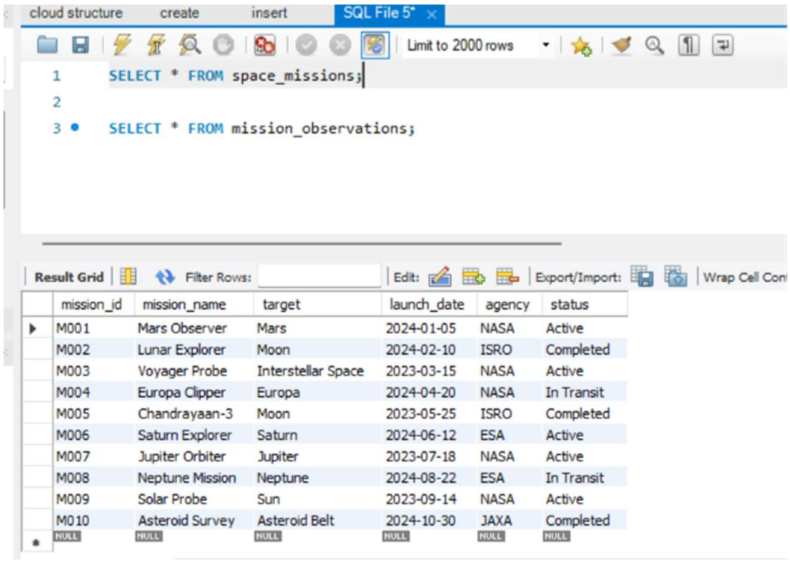
SQL Concepts Used: SELECT, COUNT(), and column aliases using AS.

Verification Performed:

- Retrieved all records from the space_missions table.

Expected Result: The database contains 10 mission records and 10 observation records.

Finding: The data retrieval and record counts help confirm that the dataset is available for further SQL analysis



mission_id	mission_name	target	launch_date	agency	status
M001	Mars Observer	Mars	2024-01-05	NASA	Active
M002	Lunar Explorer	Moon	2024-02-10	ISRO	Completed
M003	Voyager Probe	Interstellar Space	2023-03-15	NASA	Active
M004	Europa Clipper	Europa	2024-04-20	NASA	In Transit
M005	Chandrayaan-3	Moon	2023-05-25	ISRO	Completed
M006	Saturn Explorer	Saturn	2024-06-12	ESA	Active
M007	Jupiter Orbiter	Jupiter	2023-07-18	NASA	Active
M008	Neptune Mission	Neptune	2024-08-22	ESA	In Transit
M009	Solar Probe	Sun	2023-09-14	NASA	Active
M010	Asteroid Survey	Asteroid Belt	2024-10-30	JAXA	Completed

Step 3: SQL Query 1 — Filtering and Sorting Missions

Objective: To identify active space missions launched in 2024 and display them in chronological order.

SQL Concepts Used: SELECT, WHERE, AND, date filtering, and ORDER BY ASC.

Query: The SQL statement retrieves the mission ID, mission name, target, launch date, and agency for active missions launched between January 1, 2024, and December 31, 2024.

Finding: The output identifies the active missions that satisfy the specified conditions and shows their launch sequence.

```sql
SELECT
    mission_id,
    mission_name,
    target,
    launch_date,
    agency
FROM space_missions
WHERE status = 'Active'
  AND launch_date >= '2024-01-01'
  AND launch_date < '2025-01-01'
ORDER BY launch_date ASC;
```

Result Grid | Filter Rows: | Edit: | Export/Impo

mission_id	mission_name	target	launch_date	agency
M001	Mars Observer	Mars	2024-01-05	NASA
M006	Saturn Explorer	Saturn	2024-06-12	ESA
NULL	NULL	NULL	NULL	NULL

Query 2: Regional Observation Analysis

Objective: To compare space observations across different regions and identify how much data was collected in each region.

SQL Concepts Used: GROUP BY, COUNT(), AVG(), SUM(), ROUND(), and ORDER BY DESC.

Purpose: The query groups observations by region and calculates the number of observations, average signal strength, and total data collected in each region.

Finding: The results help identify regions with higher data collection and compare their average signal strength. The region with the highest total data collected appears at the top of the output.

```sql
SELECT
    region,
    COUNT(*) AS observation_count,
    ROUND(AVG(signal_strength), 2) AS avg_signal_strength,
    SUM(data_collected_gb) AS total_data_gb
FROM mission_observations
GROUP BY region
ORDER BY total_data_gb DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

region	observation_count	avg_signal_strength	total_data_gb
Sun	1	105.00	22.50
Moon	2	93.50	18.30
Jupiter	1	80.00	18.30
Jupiter System	1	76.00	15.40
Mars	1	85.00	12.50
Saturn	1	68.00	7.60
Asteroid Belt	1	72.00	6.40
Outer Solar System	2	42.00	6.00

Query 3: Combining Mission and Observation Data Using JOIN

Objective: To combine mission details with their corresponding observation records and compare their data collection.

SQL Concepts Used: INNER JOIN, table aliases, ON, and ORDER BY DESC.

Purpose: The query joins the space_missions and mission_observations tables using their common mission_id column. It displays the mission name, agency, target, observation region, distance, and data collected.

Finding: The results show how observation data relates to individual missions. Sorting by data collected makes it easy to identify the missions with the highest data collection.

```
2    SELECT
3        m.mission_name,
4        m.agency,
5        m.target,
6        o.region,
7        o.distance_mkm,
8        o.data_collected_gb
9    FROM space_missions AS m
10   INNER JOIN mission_observations AS o
11       ON m.mission_id = o.mission_id
12   ORDER BY o.data_collected_gb DESC;
13
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mission_name	agency	target	region	distance_mkm	data_collected_gb
Solar Probe	NASA	Sun	Sun	0.700	22.50
Jupiter Orbiter	NASA	Jupiter	Jupiter	778.500	18.30
Europa Clipper	NASA	Europa	Jupiter System	628.000	15.40
Mars Observer	NASA	Mars	Mars	225.400	12.50
Chandrayaan-3	ISRO	Moon	Moon	0.384	10.10
Lunar Explorer	ISRO	Moon	Moon	0.384	8.20
Saturn Explorer	ESA	Saturn	Saturn	1427.000	7.60
Asteroid Survey	JAXA	Asteroid Belt	Asteroid Belt	35.600	6.40
Neptune Mission	ESA	Neptune	Outer Solar System	4500.000	3.20
Voyager Probe	NASA	Interstellar Space	Outer Solar System	24000.000	2.80

Query 4: Identifying Above-Average Data Collection

Objective: To identify missions whose data collection exceeds the overall average.

SQL Concepts Used: Subquery, AVG(), WHERE, INNER JOIN, and ORDER BY DESC.

Purpose: The inner query calculates the average data collected across all observation records. The outer query selects missions whose data collection is greater than that average.

Finding: The results identify missions with above-average data collection. This helps compare mission performance against the dataset's overall average.

```
2    SELECT
3        m.mission_name,
4        m.target,
5        m.agency,
6        o.data_collected_gb
7    FROM space_missions AS m
8    INNER JOIN mission_observations AS o
9        ON m.mission_id = o.mission_id
10   WHERE o.data_collected_gb > (
11       SELECT AVG(data_collected_gb)
12       FROM mission_observations
13   )
14   ORDER BY o.data_collected_gb DESC;
15
```

Result Grid | Filter Rows: | Export: | Wrap Cell

mission_name	target	agency	data_collected_gb
Solar Probe	Sun	NASA	22.50
Jupiter Orbiter	Jupiter	NASA	18.30
Europa Clipper	Europa	NASA	15.40
Mars Observer	Mars	NASA	12.50

Conclusion :

The space-mission dataset was stored in MySQL using two related tables: space_missions and mission_observations. Four SQL queries were used to explore and analyse the data.

The queries demonstrated filtering and sorting, grouping and aggregate calculations, joining related tables, and using a subquery to compare data collection against the overall average. These operations helped identify active missions launched in 2024, compare observations across regions, relate mission details to observation records, and identify missions with above-average data collection.

Overall, the task provided practical experience in database querying and extracting useful information from structured data.